Rewarding Learning

**ADVANCED**
**General Certificate of Education**
**2025**

**Centre Number**

**Candidate Number**

# Digital Technology

## Assessment Unit A2 1
*assessing*
## Information Systems

*ADT11*

## [ADT11]
## FRIDAY 13 JUNE, AFTERNOON

**TIME**

2 hours 30 minutes.

**INSTRUCTIONS TO CANDIDATES**

Write your Centre Number and Candidate Number in the spaces provided at the top of this page.
**You must answer the questions in the spaces provided.**
**Do not write outside the boxed area on each page or on blank pages.**
**You are not permitted to use any calculating aid in this paper.**
Complete in black ink only. **Do not write with a gel pen.**
Answer **all six** questions.

**INFORMATION FOR CANDIDATES**

The total mark for this paper is 150.
Figures in brackets printed down the right-hand side of pages indicate the marks awarded to each question or part question.
Quality of written communication will be assessed in Questions **1(d)**, **2(c)**, **4(d)(ii)** and **5(d)**.

**1** **(a)** Describe the purpose of the following network resources.

Network card

Server

Repeater

[9]

**(b)** Describe the main features of a peer-to-peer (P2P) network.

[4]

**(c)** Explain how usernames and passwords can improve security on a server-based network.

[4]

**[Turn over**

**(d)** Bus and star are two common network topologies.

Evaluate these two topologies with respect to data security.

**Quality of written communication will be assessed in this question.**

[6]

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**(Questions continue overleaf)**

**[Turn over**

**2** **(a)** The Open Systems Interconnection (OSI) model consists of seven layers arranged in two sets.

**(i)** Distinguish between the purpose of the Application set and the Transport set.

[2]

**(ii)** Describe the main functions of each of the following layers.

The Session layer

The Network layer

[6]

**(iii)** Explain how the OSI model provides users with a wider choice of hardware and software components.

[4]

**[Turn over**

**(b)** Describe each of the following communication protocols.

CSMA/CD

RFID

[8]

**(c)** Errors can occur when data is transmitted over a computer system.

Consider the following errors.

**Error One:** The byte **$00000111_2$** is received as **$00000011_2$**

**Error Two:** The byte **$00000111_2$** is received as **$00000001_2$**

The byte contains a parity bit.

By referring to these two errors, evaluate the effectiveness of a parity bit in detecting transmission errors.

**Quality of written communication will be assessed in this question.**

[6]

**[Turn over**

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**3** **(a)** Describe the main features of a relational database.

[4]

**(b)** Explain each of the following database terms.

Entity

Primary key

[4]

**[Turn over**

**(c)** A garage keeps details about car repair bookings in a table. An extract from this table is shown below.

| BookingID | Date | Time | MechanicID | MechName | CarReg | CarType | CustName |
|---|---|---|---|---|---|---|---|
| B134 | 12/09/25 | 0930 | 1 | Black | ABC123 | Toyota | Green |
| B238 | 19/09/25 | 1100 | 4 | Grey | XYZ321 | Ford | White |
| B238 | 19/09/25 | 1100 | 2 | Brown | XYZ231 | Ford | White |
| B455 | 20/10/25 | 0930 | 3 | Brown | ABC123 | Toyota | Green |

**(i)** Normalise the data in this table to 1NF.

[2]

**(ii)** Normalise the data in this table to 2NF.

[3]

**(iii)** Normalise the data in this table to 3NF.

[4]

**(iv)** Produce an entity-relationship (ER) model for the data in the table. It should not contain any many-to-many relationships.

[5]

**[Turn over**

**(d)** A company stores the following information about its staff in a table called **Employee**. An extract from this table is shown below.

| EmployeeID | EmployeeName | Position | Department |
|---|---|---|---|
| 652 | Brown | Junior Programmer | Support |
| 655 | Green | Systems Analyst | IT |

**(i)** Produce the SQL command to add the following employee to this table.

**EmployeeID**: 720
**EmployeeName**: White
**Position**: Tester
**Department**: IT

[4]

**(ii)** Produce the SQL command to remove the employee whose **EmployeeID** is 655.

[3]

**(iii)** **EmployeeID** 652 is promoted to Senior Programmer. Produce the SQL command to make this change to the table.

_____

_____

_____

_____ [5]

**[Turn over**

**4** **(a)** Describe how the Turing Test can be used to test for artificial intelligence.

[4]

**(b)** Describe the main features of a neural network.

[4]

**(c)** Explain the purpose of the inference engine in an expert system.

[4]

**[Turn over**

**(d)** An expert system can be used to assist the diagnosis and treatment of a range of medical conditions.

**(i)** Describe the role of each of the following in the development of this expert system.

Medical consultants

An expert system shell

The knowledge engineer

[9]

**(ii)** Evaluate the use of expert systems in medicine.

**Quality of written communication will be assessed in this question.**

[6]

**[Turn over**

**5** **(a)** Describe how each of the following is used in mobile phone communication.

Cells

Handoffs

[6]

**(b)** Explain what is meant by the term robotics.

[4]

**(c)** Describe three advantages of using robots instead of humans in exploring hostile environments on Earth.

1.

2.

3.

[6]

**[Turn over**

**(d)** An organisation decides to use cloud computing for services previously provided by a LAN.

Evaluate the use of cloud computing with respect to data security.

**Quality of written communication will be assessed in this question.**

[6]

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**(Questions continue overleaf)**

**[Turn over**

**6** **(a)** Organisations using digital technology must comply with Computer Misuse legislation.

**(i)** The Computer Misuse Act states that unauthorised access to computer material is an offence. State two other offences identified in the Computer Misuse Act.

1. _____

_____

_____

_____

2. _____

_____

_____

_____ [4]

**(ii)** Describe the impact of Computer Misuse legislation on an organisation.

[6]

**(b)** Organisations using digital technology must comply with Copyright Designs and Patents legislation.

**(i)** State the purpose of the Copyright Designs and Patents Act.

[2]

**(ii)** Discuss the impact of the Copyright Designs and Patents Act on an organisation.

[6]

**THIS IS THE END OF THE QUESTION PAPER**

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| For Examiner's use only | |
|---|---|
| **Question Number** | **Marks** |
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |
| 6 | |
| **Total Marks** | |

**Examiner Number**

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